

ALTDORF, Switzerland

WMO: 066720

Lat: 46.8869N

Lon: 8.6217E

Elev: 451

StdP: 96.02

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.7	(c) -5.0	(d) -11.3	(e) 1.5	(f) -3.4	(g) -9.4	(h) 1.8	(i) -2.5	(j) 15.8	(k) 11.2	(l) 14.1	(m) 10.8	(n) 2.0	(o) 140	(p) 0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.0	(c) 29.0	(d) 19.6	(e) 27.3	(f) 19.1	(g) 25.7	(h) 18.5	(i) 20.9	(j) 27.2	(k) 20.0	(l) 25.8	(m) 19.2	(n) 24.5	(o) 4.9	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	14.2	24.2	17.8	13.5	23.1	17.1	12.9	22.3	62.3	27.2	59.2	25.9	56.5	24.7	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	10.7	8.3	-9.5	31.9	2.3	1.8	-11.2	33.2	-12.6	34.3	-13.9	35.3	-15.6	36.6
			-10.0	22.7	2.2	0.9	-11.6	23.3	-12.9	23.9	-14.2	24.4	-15.8	25.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	1.7	2.7	6.3	10.3	14.4	17.8	19.3	18.9	15.0	11.2	6.0	2.5
	DBStd	7.28	3.91	3.94	3.77	3.97	3.68	3.66	3.04	2.99	3.09	3.50	4.16	3.85
	HDD10.0	1033	258	207	123	44	6	1	0	0	2	27	132	234
	HDD18.3	2995	515	439	372	243	129	54	24	29	107	223	369	491
	CDD10.0	1237	1	2	10	53	143	234	289	275	152	63	13	2
	CDD18.3	156	0	0	0	1	9	37	56	46	7	1	0	0
	CDH23.3	1014	0	0	0	3	67	272	358	279	31	4	0	0
	CDH26.7	213	0	0	0	0	8	59	81	63	1	1	0	0
Wind	WSAvg	2.4	2.4	2.6	2.7	3.2	2.8	2.3	2.2	1.9	2.0	2.3	2.4	2.4
Precipitation	PrecAvg	1677	94	105	112	123	173	182	186	191	146	112	135	117
	PrecMax	2112	221	321	243	251	310	298	354	383	248	213	417	296
	PrecMin	1247	5	32	43	23	46	92	77	54	57	6	10	2
	PrecStd	192	58	65	52	51	77	50	65	70	50	51	80	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	15.7	18.8	23.7	27.9	30.9	31.1	31.1	26.2	23.5	19.6	17.1
		MCWB	7.6	7.3	10.3	13.5	17.0	20.1	20.7	20.6	17.4	13.6	10.8	8.8
	2%	DB	12.4	13.1	16.6	21.2	25.1	28.4	28.9	28.3	24.0	21.2	18.0	13.7
		MCWB	5.6	6.2	9.3	12.3	15.8	19.4	20.0	19.8	17.0	12.9	9.9	6.6
	5%	DB	9.6	10.8	14.8	19.3	23.3	26.4	27.0	26.3	22.2	18.6	15.2	9.9
		MCWB	5.7	5.3	8.4	11.3	14.8	18.6	19.5	19.5	16.2	12.8	8.6	6.3
	10%	DB	7.2	8.5	12.9	17.5	21.3	24.6	25.2	24.5	20.5	16.7	11.7	7.7
		MCWB	4.9	5.0	7.8	10.4	13.9	17.8	18.8	18.8	15.9	12.6	9.0	5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	8.4	11.5	14.7	18.5	21.7	22.1	21.9	19.3	15.7	11.7	9.1
		MCDB	12.2	12.5	16.7	22.0	25.7	28.6	29.2	28.3	23.9	18.7	16.3	15.0
	2%	WB	7.1	7.1	10.1	13.1	16.8	20.3	20.8	20.8	18.1	14.7	10.6	7.9
		MCDB	10.2	11.4	15.1	19.6	22.8	26.8	27.1	26.7	22.4	17.7	15.0	11.4
	5%	WB	5.9	6.1	9.1	12.0	15.6	19.3	20.0	19.8	17.1	13.9	9.9	6.8
		MCDB	8.8	9.9	13.6	17.9	21.4	25.2	25.8	25.1	20.8	17.3	14.1	9.4
	10%	WB	4.9	5.1	8.2	11.0	14.6	18.3	19.2	19.0	16.2	13.1	8.9	5.6
		MCDB	7.2	8.2	12.1	16.2	20.0	23.6	24.4	23.7	19.5	16.2	12.0	7.8
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.7	8.4	9.2	9.4	9.2	9.0	8.7	8.0	7.4	6.1	5.4
		MCDBR	9.0	10.3	11.6	12.2	13.0	12.6	12.2	11.8	10.8	10.1	9.7	8.6
		MCWBR	5.0	5.6	6.4	6.3	6.3	6.3	5.9	5.9	5.6	4.8	4.3	4.6
	5% WB	MCDBR	7.6	8.8	10.6	11.9	11.9	11.8	11.1	10.9	9.8	8.5	7.9	7.3
		MCWBR	4.9	5.5	6.3	6.7	6.2	6.2	5.7	5.7	5.5	4.8	4.5	4.6
Clear-Sky Solar Irradiance	taub	0.297	0.331	0.390	0.445	0.467	0.475	0.475	0.460	0.420	0.382	0.332	0.293	
	taud	2.284	2.185	2.088	2.022	2.046	2.088	2.098	2.151	2.234	2.299	2.347	2.317	
	Ebn at Noon	781	820	818	807	803	798	792	786	783	755	732	741	
	Edh at Noon	77	106	136	160	162	157	153	140	118	95	74	67	
All-Sky Solar Radiation	RadAvg	1.43	2.35	3.56	4.41	4.86	5.66	5.58	4.85	3.73	2.42	1.45	1.19	
	RadStd	0.17	0.29	0.50	0.69	0.49	0.51	0.49	0.46	0.37	0.28	0.20	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	+0.51	N/A	N/A	N/A	N/A
Regional (117 neighbors)	N/A	N/A	N/A	+0.72	+0.41	+0.32	N/A	N/A	+57	+15

Nomenclature: See separate page